

BIOLOGICAL SCIENCE Part 12

1. Which statement/s is /are TRUE of adaptation?

- I. An adaptation
 - II. It is genetic change
 - III. Due to the help nature of the mutation, it is passed down from one generation to the next
 - IV. Mutation is inherited and so becomes an adaptation
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- A. I and II only
 - B. II and III only
 - C. I, II and III only
 - D. I, II, III and IV

Ans: An adaptation is a mutation, or genetic change, that helps an organism such as a plant or animal, survives in its environment. Due to the helpful nature of the mutation, it is passed down from one generation to the next. As more organism inherit the mutation becomes a typical part of the species. The mutation has become an adaptation.

2. Which of the following is NOT an example of behavioral adaptation?

- A. Fur on a bear
- B. Migration of birds
- C. Opposum playing dead
- D. Rabbit freezing when it thinks is has been seen.

Ans: A. Fur on a bear

-All organism have adaptation that help them survive and thrive. Some adaptation are structural. Structural adaptations are physical features of an organism like bill on a bird or the fur on a bear. Other adaptations are behavioral. Behavioral adaptation are the things organism do to survive. For example, bird calls and migration, opossum plays dead and rabbit freezes when seen are behavioral adaptations

3. A chameleon changing colors to blend in with its surroundings is an example of _____

- A. camouflage
- B. extinction
- C. Hibernation
- D. migration

Ans: A camouflage

- Chameleons are capable of changing color. These colors changes assist the reptile to camouflage itself in the presence of potential predators. Chameleons tend to move very slowly. They remain perfectly still into the surroundings vegetation, making it hard for

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predators to spot them. Color changes also help the chameleon regulate its body temperature, as darker body colors absorb heat and lighter colors reflect and thus repel heat.

4. Which process do some animals undergo when they go into a deep winter sleep and so body activities slowdown and can live off store food?

- A. camouflage
- B. extinction
- C. Hibernation
- D. migration

Ans: C. Hibernation

- Groundhogs also called woodchucks, spend the warmer months eating to build up the fat layers to survive a winter in hibernation. Hibernation is a deep sleep that helps them to save energy and survive the winter without eating much. During hibernation the animal's body temperature drops and its heartbeat and its breathing slowdown so that it does not use much energy.

5. Which adaptation is employed by a viceroy butterfly when it copies the monarch butterfly so that it does not get eaten?

- A. Camouflage
- B. Mimicry
- C. Play dead
- D. Venom

Ans: B. Mimicry

- Mimicry comes in several varieties, including Batesian mimicry, which is when a palatable organism mimics a species that is unpalatable to predators. Consequently, they are avoided by predators increasing their fitness. A vivid example of Batesian mimicry is depicted by Viceroy and monarch Butterflies. Monarch Butterflies are unpalatable due to toxic milkweeds they consume as larvae, which results in low levels of predation in their natural environment.

6. There is unity in diversity in the study of biology. Which statement explains this?

- A. All the living beings have structures unique to each
- B. Living beings have structures unique to each
- C. No living being can live alone
- D. Living things need one another

Ans: A. All the living beings have structures unique to each

- The meaning of unity in diversity is the existence of oneness even after various differences. As diverse as life is, it also displays remarkable unity there are genetic

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similarities within the DNA of organism as different as bacteria and animals. Unity is also evident in many features of cell structure.

7. Which explains/s complementary of structure and function?

- I. Function always reflects structure
- II. The physiology of a body parts is determined by its anatomy
- III. The function of a body part depends on its structure

- A. I, II and III
- B. I and II
- C. II and III
- D. I and III

Ans: A. I, II and III

- The principle of complementarity of structure and function states that function is dependent on structure and that the form of a structure relates to its function.

8. The sharp incisors (The teeth in front) can tear food, while the molars (the flat teeth in back) can grind food into tiny pieces. The molars are also closest to the higher of the Jaw so that's where chewing can occur with the most force especially useful for grinding up harder-to-digest plant foods. What does this proves ?

- A. Function is dependent on structure.
- B. Structure does not relate to its function
- C. Structure and functions evolve
- D. Structure and function are poles apart.

Ans: A. Function is dependent on structure.

- The principle of complementarity of structure and function states that function is dependent on structure and that the form of a structure relates to its function. Bone are strong and contain hard mineral deposits so they can support the body.

9. Which of the following is TRUE?

- A. A molecule consist of two or more atoms of the same different elements
- B. Molecules fuse together to form atoms which come together to form electrons
- C. Electrons and protons combine to form molecules, which come together to form atoms
- D. Protons are the fundamental structural units of elements and each proton retains all the properties of its element

Ans: A. A molecule consist of two or more atoms of the same different elements

- A molecule is composed of two or more atoms of the same or different elements, held together by interactions among their outermost electron shells. Meanwhile an atom has subatomic particles which include protons, neutrons and electrons

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10. Which of the following is NOT an important electrolyte in the body?

- A. Carbon ions
- B. Chloride ions
- C. Potassium ions
- D. Sodium ions

Ans: Electroare eletrically-charged minerals the body require for regulating water levels blood acidity and muscle function. The most common electrolytes are sodium, potassium and chloride; other electrolytes include calcium, magnesium and phosphorus. Sodium positive in the body determines the volume of extracellular fluid in circulation, making it cirital for maintaining blood pressure. Chloride, a negatively charge ion, influences neural activity and muscle function. Potassium plays a vital role in cellular and electrical functions in the body.

11. What is an atomic mass?

- A. It is sum of electrons and protons
- B. It is sum of electrons and neutrons
- C. It is the total number of protons and neutrons
- D. It is difference in the number of protons and neutrons

Ans: C. It is the total number of protons and neutrons

- The atomic mass of an element is the total number of protons and neutrons. It is the measure of the atom's size and has a key role in the chemical properties of the elements. Although number is the number of protons in the nucleus.

12. Glucose is the carbohydrate and is considered to be the most important simple sugar in human metabolism. It has a chemical formula of $C_6H_{12}O_6$. What does this mean?

- A. It contains 6 atoms of carbon 12 atoms and oxygen and 12 protons of hydrogen.
- B. It contains 6 atoms of carbon 12 atoms of hydrogen and 6 atoms of oxygen
- C. It contains 6 molecules of carbon 12 molecules of hydrogen and 6 molecules or oxygen
- D. The atomic mass of hydrogen is equal to the sum of the atomic masses of carbon and oxygen

Ans: B. It contains 6 atoms of carbon 12 atoms of hydrogen and 6 atoms of oxygen

- Glucose is a simpler sugar. It is a compound with a molecular formula of $C_6H_{12}O_6$. Since it is a compound, it is describes in terms of its constituent atoms.

13. How many atoms of each elements are there in $2Al_2(SO_4)_3$?

- A. 2 atoms of aluminum, 3 atoms of sulfur and 7 atoms of oxygen
- B. 4 atoms of aluminum. 3 atoms of sulfur and 7 atoms of oxygen

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- C. 4 atoms of aluminum, 3 atoms of sulfur and 12 atoms of oxygen
- D. 4 atoms of aluminum. 6 atoms of sulfur and 24 atoms of oxygen

Ans: D. 4 atoms of aluminum. 6 atoms of sulfur and 24 atoms of oxygen
- $(\text{SO}_4)_3$ would mean that there are three ($1 \times 3 = 3$) sulfur atoms and twelve ($4 \times 3 = 12$) oxygen atoms. The coefficient of 2 in front indicates that there are two $\text{Al}_2(\text{SO}_4)_3$ molecules. Therefore, in total, there are four aluminum atoms, six sulfur atoms and 24 oxygen atoms.

14. In the chemical equation $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} \downarrow + \text{NaNO}_3$, what does $\downarrow$ mean?

- A. This is an aqueous solution
- B. Gas evolves
- C. Heat is supplied
- D. Precipitate is formed

Ans: D. Precipitate is formed

-The symbol (aq) means aqueous. Means heat is supplied means evolution of gas means formation of precipitate.

15. Where can you find the non-metals in the periodic table?

- A. Bottom part
- B. left side
- C. Middle portion
- D. Right side

Ans: D. Right side

- Metals are usually found at the left side up to the middle part of the periodic table. Elements on the right side are classified as non-metals. Metalloids are the elements in between metals and non-metals.

16. Oxygen, an important gas in respiration, belongs to Group VI-A in the periodic table does the number in the group indicate?

- A. Energy level
- B. Number of protons
- C. Oxidation number
- D. Valence electrons

Ans: D. Valence electrons

- Group is the column in the periodic table. The elements in each group have the same number of electrons in the outer orbital. Those outer electrons are known as valence electrons. They are the electrons participating in chemical bonding.

17. How is the modern periodic table arranged?

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- A. Increasing atomic mass
- B. Based on the year discovered
- C. Increasing atomic number
- D. Decreasing electronegativity

Ans: A. Increasing atomic mass

- The periodic table is a tabular arrangement of elements based on increasing atomic number. The modern periodic law states that the properties of elements and their compounds are a periodic function of their atomic numbers.

18. Which of the chemical bond plays a key role in shaping tertiary and quaternary proteins in determining shapes of chromosomes and in muscle contraction?

- A. Covalent
- B. Hydrogen
- C. Ionic
- D. Polar covalent

Ans: C. Ionic

- Ionic bonds are commonly found in dry forms such as salts are found in compounds throughout the body. Ionic bonds play a major in shaping tertiary and quaternary proteins in determining the shapes of chromosomes depending on what atoms bond to each other, in cell signaling and in muscle contraction. Most biological molecules employ covalent bonding. Monosaccharides such as glucose, fructose and galactose are held together amino acids in chain creating peptides. The double strands of DNA are held together by hydrogen bonds.

19. Which chemical bonding is involved in the formation of water?

- A. Covalent
- B. Hydrogen
- C. Ionic
- D. Polar covalent

Ans: D. Polar covalent

- In water, an electron is shared between each hydrogen atom and the central oxygen atom. The oxygen atom attracts the electron a little more strongly than hydrogen does this means. That the shared electrons spend more time at the oxygen end of the molecule slightly negative. Since the electrons are not near the hydrogen end as much that end is slightly positive. When a covalently bonded molecule has more electrons in one area than another, it is called a polar molecule. Water is an example of a polar molecule.

20. The molecule structure of carbon dioxide is $\text{O}=\text{C}=\text{O}$. What does this tell you?

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- A. It is formed through ionic bonding
- B. It has a stronger bond compared to carbon monoxide
- C. It is composed of two atoms of oxygen have double bond each between their atoms
- D. Carbon and oxygen have doubled bond each between their atoms

Ans: D. Carbon and oxygen have doubled bond each between their atoms

-The Octet Rule requires all atoms in a molecule to have 8 valence electrons either sharing losing or gaining electrons to become stable. Carbon dioxide has a total of 1 carbon atom and 2 oxygen atoms. Each oxygen atom has 6 valence electrons whereas the carbon atom only has 4 valence electrons. Satisfy the Octet Rule, carbon needs 4 more valence electrons. Since each oxygen atom has 3 lone pairs of electrons, they can each share 1 pair of electrons with carbon; as a result, filling carbon's outer valence shell thereby satisfying the Octet Rule. The molecular structure of the result shows a double bond. A double bond is when two atoms share two pairs of electron with each other. The double bond is depicted conventionally by two joining the atoms. Each line represents one pair of shared electrons. This type of bond is much stronger than a single bond, but less stable.

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